



Energy Management System

Real-time monitoring, intelligent
control, and proof of savings



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Visibility

Energy use & equipment health
live Dashboards, Customizable
Energy Reports & Audit Reports



Control

Adjust Setpoints Chiller, AHU ,
FCU, HVLS & Lighting centrally
Trends, Events & Alarms Cloud
Connectivity



Governance

Zone based control Times Schedules
Security logs & role-based access AI
powered

Why Zealcorps EMS

Most buildings lose money in two places:

- Running equipment when it is not needed
- Not spotting issues early.
- In addition, Zealcorps applies a unique AI-driven optimization strategy that converts the BMS into a mathematical model and produces advisory recommendations for clients to enhance efficiency, reliability, and overall energy performance.

Visibility

Lower bills through better operating hours and fewer unnecessary run-times

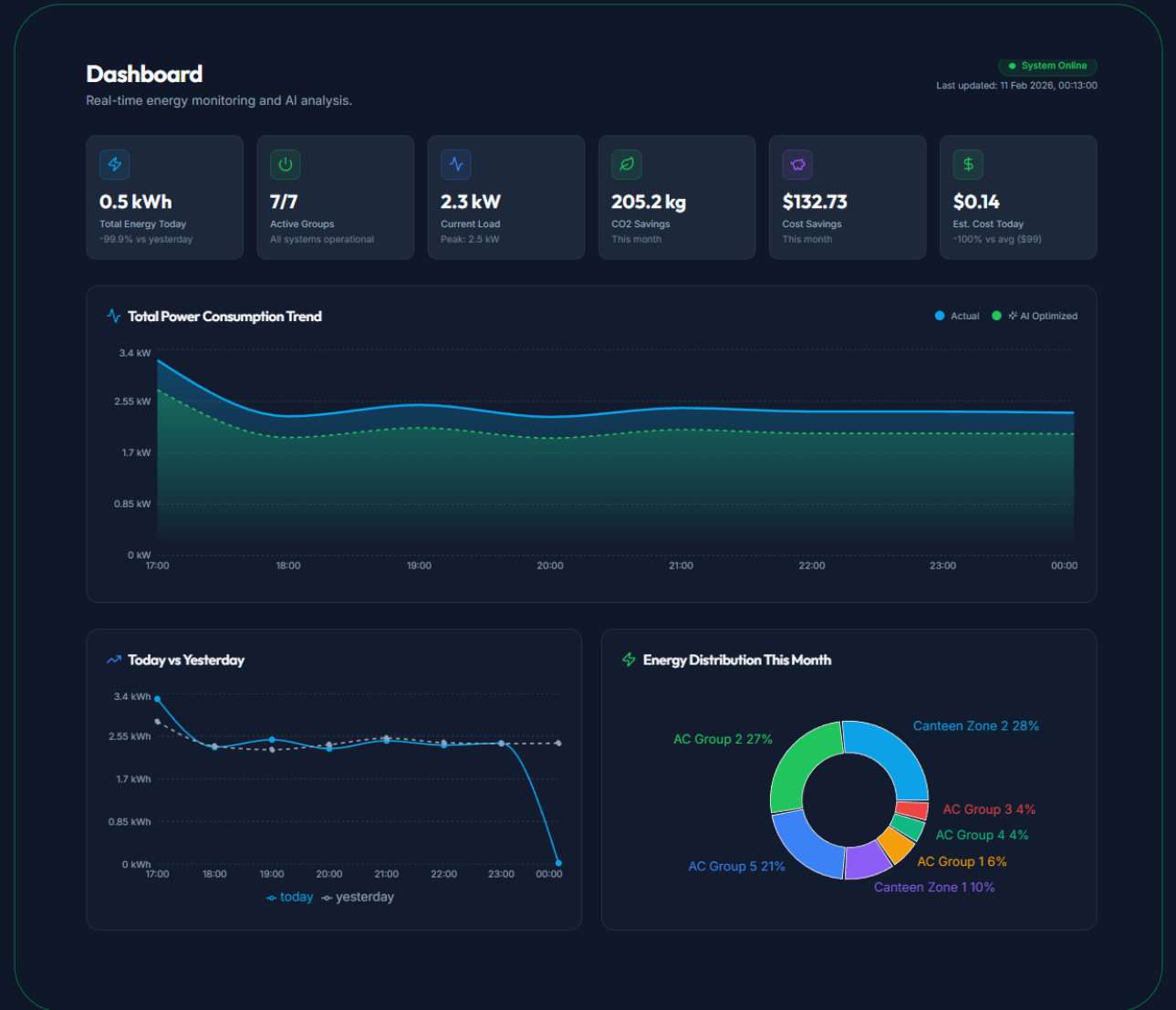
More comfortable spaces (temperature and humidity are visible and trackable)

Faster troubleshooting: live status and logs reduce guesswork

Audit-ready exports (CSV/PDF) for management, reporting, and compliance

Dashboard: real-time monitoring and AI analysis

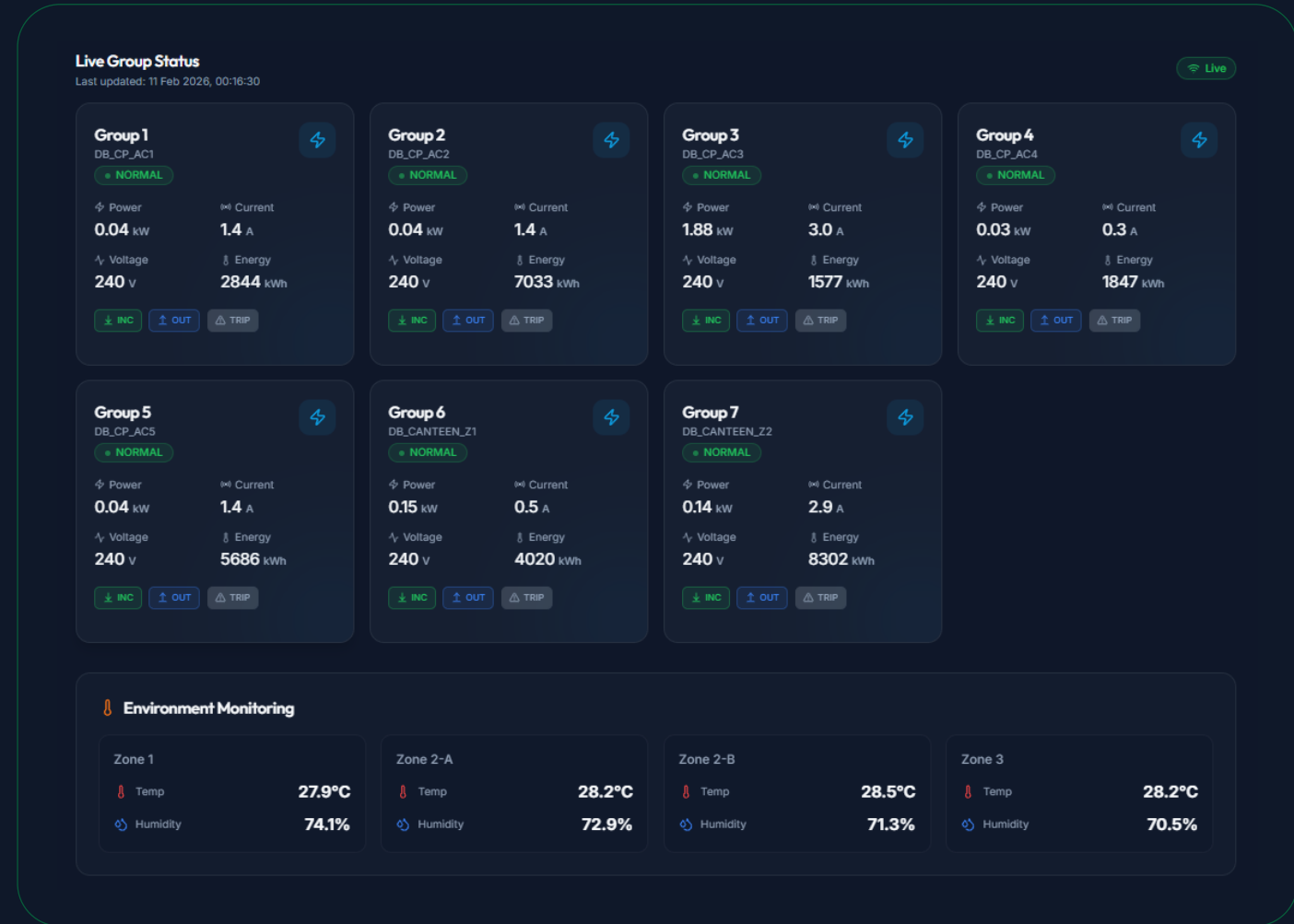
- Initial Automation
- Occupancy Sensors Added
- Zone Control Implementation
- Predictive AI Models
- Weather-Adaptive Systems
- Cloud AI Integration



Live Group Status: Operational impact

Measure & verify

- Higher kWh and peak demand charges
- Unplanned alarms and manual intervention
- Comfort / SLA risk when load changes quickly
- Hard to prove results to management



Live Group Status: Operational impact

Dynamic setpoints

Imagine driving a car with cruise control:

Static: You always drive at 100 km/h no matter traffic, hills, or weather.

Dynamic: Your speed automatically adjusts based on traffic, road slope, and fuel efficiency.

Dynamic setpoints do the same thing for energy systems.

Control Settings

Manage and control individual units across FCU, Lighting, and HVLS systems

FCU

Lighting

HVLS

FCU (Fan Coil Unit)

0 of 33 units active across 3 zone(s)

Last updated: 11 Feb 2026, 00:16:45

Live

Zone 1

0 of 7 units active

All Off

Unit 0

Auto Manual

Current 27.8°C

Setpoint 23°C

16°C 30°C

Off Standby

Unit 1

Auto Manual

Current 26.6°C

Setpoint 24°C

16°C 30°C

Off Standby

Unit 2

Auto Manual

Current 26.7°C

Setpoint 24°C

16°C 30°C

Off Standby

Unit 3

Auto Manual

Current 30.4°C

Setpoint 24°C

16°C 30°C

Off Standby

Unit 4

Auto Manual

Current 30.3°C

Setpoint 24°C

16°C 30°C

Off Standby

Unit 5

Auto Manual

Current 30.6°C

Setpoint 24°C

16°C 30°C

Off Standby

Unit 6

Auto Manual

Current 30.5°C

Setpoint 24°C

16°C 30°C

Off Standby

Schedules: Smart Scheduling Groups

Schedule optimization

If demand is:

50% demand → Run Group A + partial Group B

75% demand → Run Group A + Group B

100% demand → Run all groups

Demand controls schedules. Schedules activate groups. Groups optimize efficiency.

Equipment Schedules

Configure on/off schedules for equipment sets by day

Holidays (2)

+ Add Schedule

Lighting (1 Set)

FCU (3 Sets)

HVLS (1 Set)

Lighting - Set 1

5 schedules

Day	Schedules	Actions
Monday	08:00 - 18:00  	+
Tuesday	08:00 - 18:00  	+
Wednesday	08:00 - 18:00  	+
Thursday	08:00 - 18:00  	+
Friday	08:00 - 18:00  	+
Saturday	No schedules	+
Sunday	No schedules	+
Holiday	No schedules	+

Reports: energy consumption with exportable Reports

Load forecasting

Predictive AI Models

Machine learning models started predicting thermal loads to proactively optimize energy consumption, improving efficiency.

Energy Reports

View and export energy consumption, trip events, and operation logs

Report Filters

Date Range

Feb 01, 2026 - Feb 10, 2026

Group

Total (MDB CANTEEN) ▾

Hourly

Daily

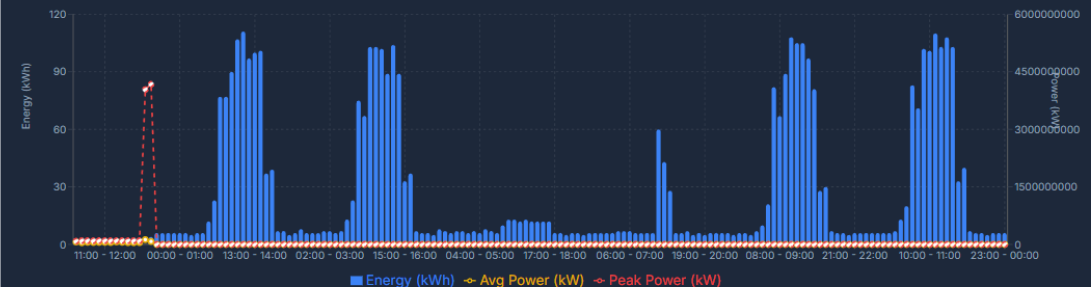
Monthly

Zone Control

Trips

Operations

Hourly Energy Consumption



Hourly Breakdown

CSV PDF

Date	Hour	Energy (kWh)	Avg Power (kW)	Peak Power (kW)	Avg Voltage (V)	Peak Voltage (V)	Avg Current (A)
04/02/2026	06:00 - 07:00	0.00	64837719.85	85300027.00	3.0	3.0	8252.85
04/02/2026	07:00 - 08:00	0.00	44088822.49	99940077.00	3.0	3.0	19160.65
04/02/2026	08:00 - 09:00	0.00	55762746.76	98900068.00	3.0	3.0	19665.58
04/02/2026	09:00 - 10:00	0.00	51365509.38	97900096.00	3.0	3.0	23709.72
04/02/2026	10:00 - 11:00	0.00	53627678.36	99940090.00	3.0	3.0	24306.44
04/02/2026	11:00 - 12:00	0.00	54812427.33	99740083.00	3.0	3.0	23795.44

Zone Control reports: Weather-Adaptive Systems

Zone Control

zone-based controls, managing lighting and HVAC independently per to optimize energy use.

Adjusting setpoints dynamically based on real-time weather data for better temperature and fan control.



AI - Optimisation : ESG = Financial Advantage

Operations logs

Immediate Cost Reduction

Running the system at its most efficient operating state — while maintaining or improving comfort and reliability.

Predictable Financial Planning

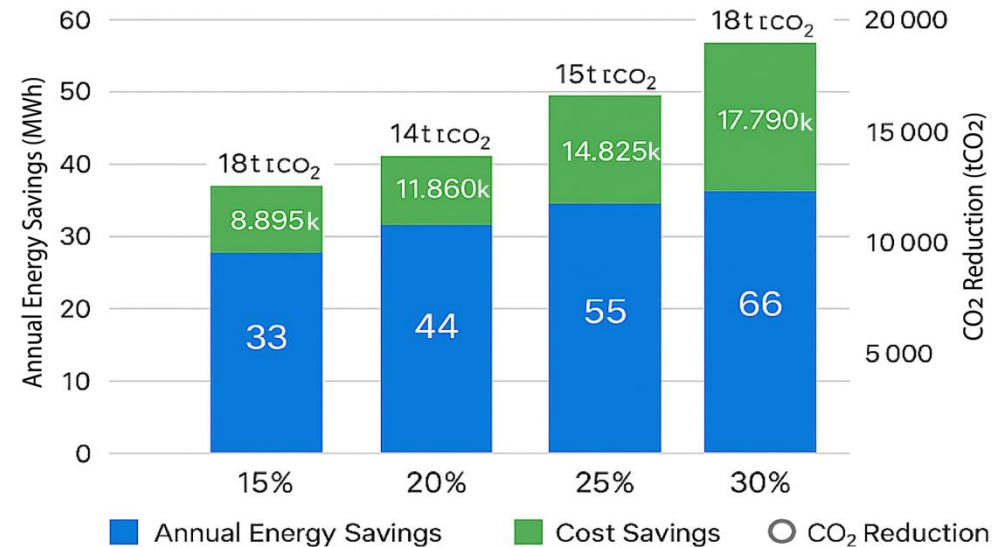
“Based on conservative financial modeling using historical utility data, the system is structured to guarantee full payback within 24 months.”



Summary of Energy Savings (Projected)

1. AI-driven time scheduling prevents unnecessary after-hours and weekend energy use, saving up to 10 %.
2. Zone-based controls tailor energy use per room type and occupancy, optimizing consumption by up to 8 %.
3. Occupancy sensors automate lighting and HVAC, eliminating waste in unoccupied spaces and saving up to 8 %.
4. Dynamic weather-adaptive adjustments ensure HVAC systems respond to real-time conditions, improving efficiency by up to 4 %.

Energy and Cost Savings at 15–30% Reduction



Saving %	Energy Saved (MWh)	Cost Saved (SGD)	CO ₂ Reduction
15 %	33	8.895k	17.7
20%	44	11.860k	23.6
25%	55	14.825k	29.5
30%	66	17.790k	35.4

Average electricity tariff = 0.27 SGD/kWh

Operations logs: Audit Trail = Accountability + Faster Investigations

AI Optimization Impact

Immediate Cost Reduction

Running the system at its most efficient operating state – while maintaining or improving comfort and reliability.

Predictable Financial Planning

“Based on conservative financial modeling using historical utility data, the system is structured to guarantee full payback within 24 months.”

Energy Reports

View and export energy consumption, trip events, and operation logs

Report Filters

Date Range

Feb 01, 2026 - Feb 10, 2026

Equipment Type

All Equipment

Operation

All Operations

Hourly

Daily

Monthly

Zone Control

Trips

Operations

Operation Logs (161)

CSV

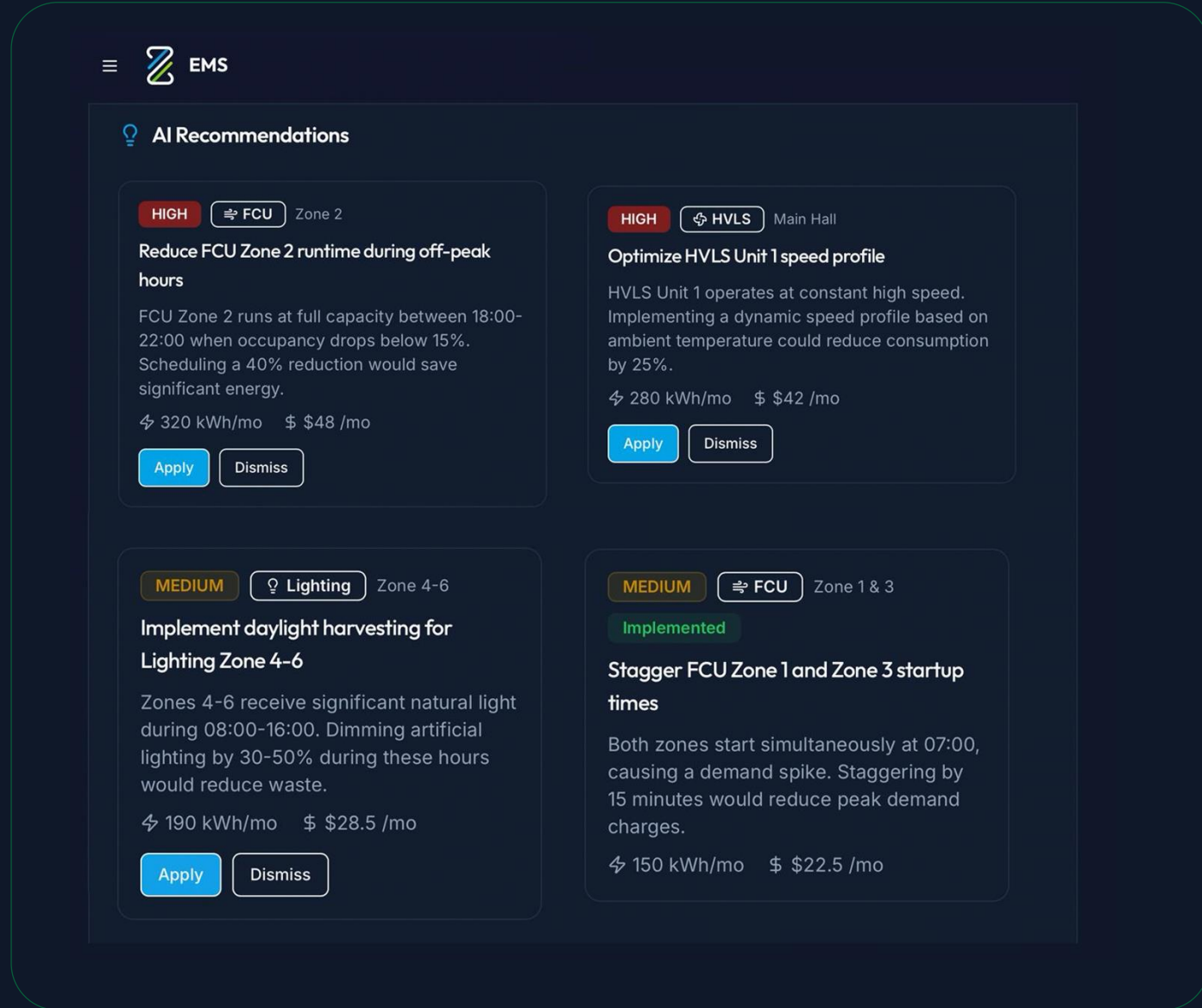
PDF

Date/Time	Equipment Type	Zone/Unit	Operation	Description	Updated By	Status	Error
10/02/2026, 06:51:12 pm	Lighting	-	Delete Schedule Slot	Lighting Set 1 schedule deleted (Friday, slot 1, 08:00-18:00)	Default Admin	Succeeded	-
10/02/2026, 06:51:11 pm	Lighting	-	Delete Schedule Slot	Lighting Set 1 schedule deleted (Thursday, slot 1, 08:00-18:00)	Default Admin	Succeeded	-
10/02/2026, 06:51:09 pm	Lighting	-	Delete Schedule Slot	Lighting Set 1 schedule deleted (Wednesday, slot 1, 08:00-18:00)	Default Admin	Succeeded	-
10/02/2026, 06:51:06 pm	Lighting	-	Delete Schedule Slot	Lighting Set 1 schedule deleted (Monday, slot 1, 08:00-18:00)	Default Admin	Succeeded	-
10/02/2026, 06:49:55 pm	Lighting	-	Add Schedule Slot	Lighting Set 1 schedule added (Monday, slot 1, 08:00-18:00)	Default Admin	Succeeded	-
10/02/2026, 06:49:42 pm	Lighting	-	Add Schedule Slot	Lighting Set 1 schedule added (Friday, slot 1, 08:00-18:00)	Default Admin	Succeeded	-
10/02/2026, 06:49:40 pm	Lighting	-	Add Schedule Slot	Lighting Set 1 schedule added (Thursday, slot 1, 08:00-18:00)	Default Admin	Succeeded	-
10/02/2026, 06:49:37 pm	Lighting	-	Add Schedule Slot	Lighting Set 1 schedule added (Wednesday, slot 1, 08:00-18:00)	Default Admin	Succeeded	-
10/02/2026, 06:49:31 pm	Lighting	-	Add Schedule Slot	Lighting Set 1 schedule added (Tuesday, slot 1, 08:00-18:00)	Default Admin	Succeeded	-

Intelligent Energy Recommendations

AI : Analyze + Recommend + Optimize

- AI evaluates equipment behavior and consumption trends across building systems.
- Smart alerts identify opportunities to reduce energy waste.
- Context-aware recommendations improve HVAC, lighting, and mechanical efficiency.
- Automated optimization suggestions help reduce operational costs.
- Clear visibility of potential energy and cost savings for each recommendation.
- Continuous AI learning improves prediction accuracy over time.



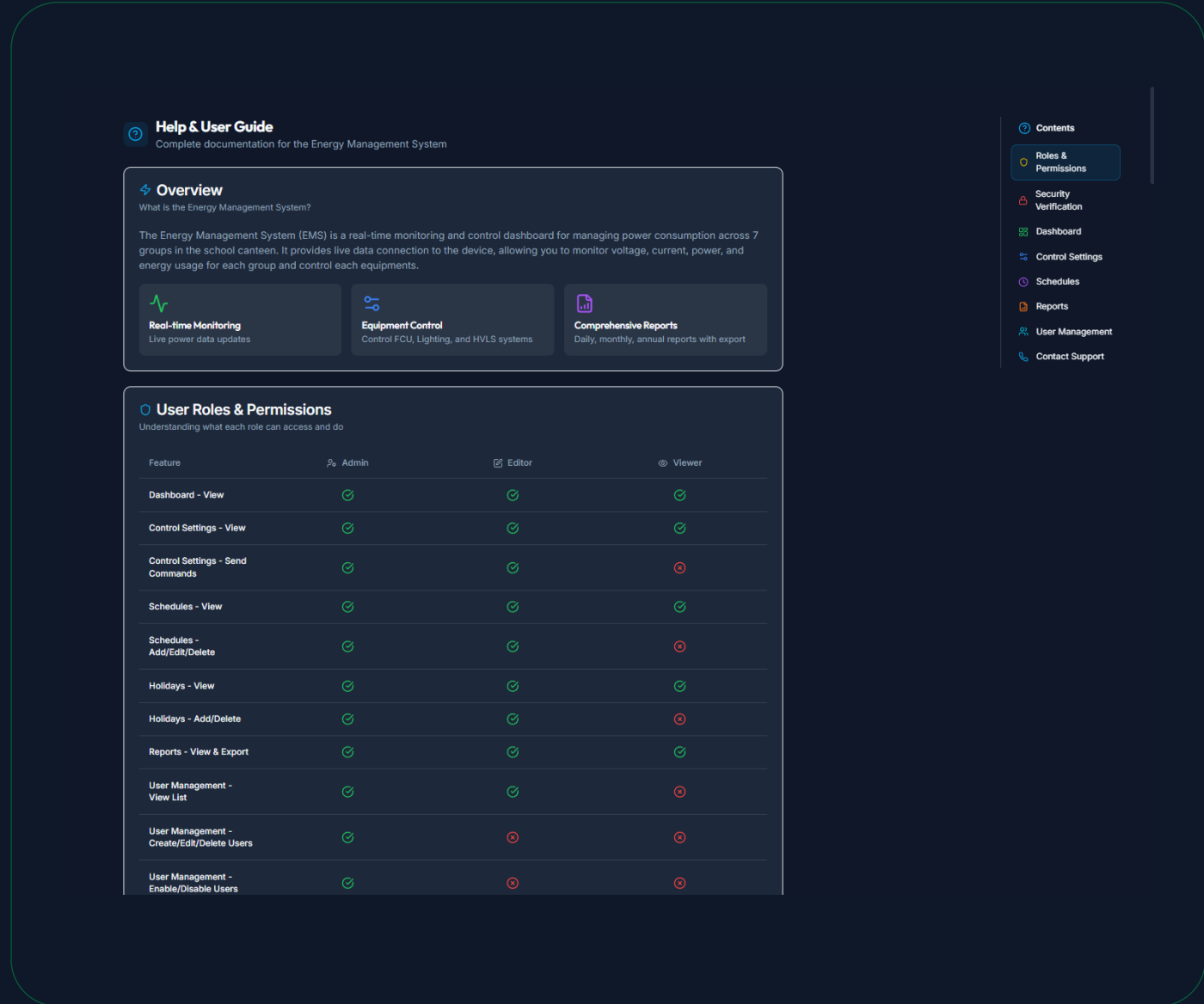
The screenshot displays the 'AI Recommendations' section of an EMS interface. It features four recommendation cards, each with a severity level (HIGH or MEDIUM), a category icon, a title, a description, potential savings, and 'Apply' or 'Dismiss' buttons.

- Recommendation 1 (HIGH):** FCU Zone 2. Title: 'Reduce FCU Zone 2 runtime during off-peak hours'. Description: 'FCU Zone 2 runs at full capacity between 18:00-22:00 when occupancy drops below 15%. Scheduling a 40% reduction would save significant energy.' Savings: 320 kWh/mo, \$48 /mo. Buttons: Apply, Dismiss.
- Recommendation 2 (HIGH):** HVLS Main Hall. Title: 'Optimize HVLS Unit 1 speed profile'. Description: 'HVLS Unit 1 operates at constant high speed. Implementing a dynamic speed profile based on ambient temperature could reduce consumption by 25%.' Savings: 280 kWh/mo, \$42 /mo. Buttons: Apply, Dismiss.
- Recommendation 3 (MEDIUM):** Lighting Zone 4-6. Title: 'Implement daylight harvesting for Lighting Zone 4-6'. Description: 'Zones 4-6 receive significant natural light during 08:00-16:00. Dimming artificial lighting by 30-50% during these hours would reduce waste.' Savings: 190 kWh/mo, \$28.5 /mo. Buttons: Apply, Dismiss.
- Recommendation 4 (MEDIUM):** FCU Zone 1 & 3. Title: 'Stagger FCU Zone 1 and Zone 3 startup times'. Description: 'Both zones start simultaneously at 07:00, causing a demand spike. Staggering by 15 minutes would reduce peak demand charges.' Savings: 150 kWh/mo, \$22.5 /mo. Status: Implemented (green bar). Buttons: Apply, Dismiss.

Roles & permissions Security : Access control + audit + OT-safe design

AI Optimization Impact

- Cloud service hardened to ISO 27001 / SOC 2 / IM8
- TLS 1.3 encryption for MQTT/API
- AES-256 encryption at rest
- MFA + RBAC + SSO for all user and API access
- Key Vault Securely stores encryption keys, credentials, and API secrets used by EMS and AI.
- Azure Front Door Acts as global secure entry point with DDoS protection and load balancing.



The screenshot displays the 'User Roles & Permissions' section of the Energy Management System (EMS) interface. The page includes a sidebar with navigation links: Contents, Roles & Permissions (active), Security Verification, Dashboard, Control Settings, Schedules, Reports, User Management, and Contact Support. The main content area is titled 'User Roles & Permissions' and includes a sub-header 'Understanding what each role can access and do'. Below this is a table with columns for Feature, Admin, Editor, and Viewer, showing the permissions for various system features.

Feature	Admin	Editor	Viewer
Dashboard - View	✓	✓	✓
Control Settings - View	✓	✓	✓
Control Settings - Send Commands	✓	✓	✗
Schedules - View	✓	✓	✓
Schedules - Add/Edit/Delete	✓	✓	✗
Holidays - View	✓	✓	✓
Holidays - Add/Delete	✓	✓	✗
Reports - View & Export	✓	✓	✓
User Management - View List	✓	✓	✗
User Management - Create/Edit/Delete Users	✓	✗	✗
User Management - Enable/Disable Users	✓	✗	✗



**Connect with us and start unlocking
greater savings with ZEMS.AI**

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